

Cell Design Specification — VBF-T1R1FLASH-DS-01

Case: next-generation BEV sedan battery (ED ≥ 392.61 Wh/kg · 4C fast charge no plating · T_{max} ≤ 60 °C · overcharge to 4.7 V no thermal runaway)

Generation date: 2026-08-25 · Source: OKane2022 parameter set + bda simulation outputs (file:key annotated per line)

1. Basic specification

Field	Value	Source
Electrochemical system	NMC811 / graphite+SiOx (OKane2022 parameterization)	entry-0 meta.base_params
Nominal capacity (Ah)	4.936 (1C DFN discharge)	cell/final_1c_dfn.json:capacity_ah
Voltage window (V)	2.5 – 4.2 OKane2022 parameter set (Lower/Upper voltage cut-off)	
Cell form	Pouch strip 1580 mm × 65 mm; total stack thickness 0.1908 mm	Electrode height/width/thickness params
Shell/casing thickness	Not provided (parameter set has no shell)	–
Electrolyte	High-transport formulation: σ = 5.0 S/m, D = 1.2e-9 m²/s, t ₊ = 0.65 (design override, estimate); base Nyman2008 σ(45 °C)=1.46 S/m, D=2.72e-10, t ₊ =0.259	candidates/archA_h150_f4b.json + pybamm parameter dump
Cation transference number	0.65 (design override)	candidates/archA_h150_f4b.json

2. Electrode and separator

Layer	Thickness (μm)	Porosity	Material	Source
Positive electrode	75.6	0.335	NMC811	OKane2022 params
Negative electrode	85.2	0.25	graphite+SiOx	OKane2022 params
Separator	12.0	0.47	polyolefin	OKane2022 params
Positive current collector	10.0 (stock 16)	–	Al	candidates/archA_h150_f4b.json
Negative current collector	8.0 (stock 12)	–	Cu	candidates/archA_h150_f4b.json
N/P	1.13 (85.2/75.6 thickness balance; capacity-caliber N/P not separately parameterized)			OKane2022 stock balance

3. Process design parameters

Parameter	Value	Formula / source
Positive areal density	164.0 g/m²	thickness×(1-ε)×density = 75.6e-6×0.665×3262; = layer_kg_m2
positive_electrode	0.163994 kg/m²	
Negative areal density	105.9 g/m²	85.2e-6×0.75×1657 = 0.105882 kg/m²
Positive compaction density	2.17 g/cm³	3262×0.665 = 2169 kg/m³ ÷1000
Negative compaction density	1.24 g/cm³	1657×0.75 = 1243 kg/m³ ÷1000
Electrolyte fill amount	6.44 g	pore volume 5.37e-6 m³ × 1.2 g/cm³ (literature density, annotated)
Formation	0.1C CC to 4.2 V, 25 °C, 2 cycles	design recommended value; production-line value requires tuning
Cooling requirement	h ≥ 120 W/m²·K (liquid cooling; design point 150)	robustness probe F4b-h120 passed

4. Mass breakdown (contract caliber, electrolyte excluded)

Layer	kg/m²	Mass (g)	Source
Positive electrode	0.163994	16.84	calc-energy layer_kg_m2 × area 0.1027 m²
Negative electrode	0.105882	10.87	same
Positive CC (Al)	0.027000	2.77	same
Negative CC (Cu)	0.071680	7.36	same
Separator	0.002525	0.26	same
Total	–	**38.11**	cell/final_energy.json:mass_kg (electrolyte excluded; +6.44 g electrolyte per BOM caliber)

5. Performance verification (vs entry-0 criteria)

Metric	Result	Threshold	Verdict	Source
Energy density	459.41 Wh/kg	≥ 392.61	✓	cell/final_energy.json:energy_density_wh_kg
4C charge T _{max}	323.39 K (50.2 °C)	≤ 333.15 K	✓	cell/archA_h150_f4b_4c.json:T_max_K
4C charge plating	anode min +0.0114 V	≥ 0 V (plated=false)	✓	cell/archA_h150_f4b_4c.json:anode_potential_v
Overcharge 4.7 V → TR	triggered = false	triggered=false	✓	cell/final_tr.json:triggered

6. Design notes (rationale from evaluate log, rounds 1–6)

- Electrolyte transport overrides (σ/D/t₊): root-cause fix for end-of-charge anode surface saturation at 4C (rounds 3–5; anode min –0.077 → –0.029 → –0.0023 → +0.0114 V). Values are literature-informed estimates for high-conductivity/single-ion-like formulations.
- Thin collectors (16/12 → 10/8 μm): ED boost 405.6 → 462.6 Wh/kg at zero electrochemical side effect (round 1).
- Liquid cooling h = 150 (robustness verified at h = 120): T_{max} fix (367.4 → 323.4 K) without ED cost (round 2); thermal-management DOF, undeclared in task → widest interpretation.
- N/P increase rejected by measurement (round 4): deeper charge into anode saturation, no plating benefit.